Exam 1 Practice

1) Make **corrections** to the following code to fix any syntax or logical errors that would prevent it from either executing or providing the intended answer. Points will be deducted for incorrect changes that "break" the code.

```
def output_BMI():
    """Computes and outputs a person's BMI (body mass index)
    """
    #compute BMI as a float
    BMI = weight * 703 // height**2

    #Output BMI with category
    if BMI < 18.5:
        print(f"BMI: {BMI:>12.1f})
    else BMI >= 18.5 and <25:
        print(f"BMI: {BMI:>12.1f}")
    else:
        print(f"BMI: {BMI:12.1f}")

def main():
    #Prompt user for input
    w = input("Please enter weight in pounds: ")
    h = input("Please enter height in inches: ")
    output_BMI(w, h)

if __name__ == "__main__":
    main()
```

2) How many 'B's are output?

```
b = "BBB"
for x in range(2, 4):
    for y in range(9, 7, -1):
        b = b + "B"
        print("B")
    print("B")
print(b)
```

3) What is the output of the following?

a)

```
x = ["Willie", "Jonas", "Roy", "Eunice"]
gilligan = x[1: ][2].find("e")
print(gilligan)
```

prints: 5

b)

```
a = 2
b = 7
while b > 3:
    for a in range(1, 5, 1):
        b = b - 2
        if a >= b:
            a = a * 2
            print(f"{a:5}", end = " ")
    print(f"{a*1000:10,}")
print(f"{b=}")
```

4) Write a function called `add_up` that is passed no parameters. The function adds positive integers entered by the user until a -1 is entered. The function returns the sum of all numbers added aside from the final -1.

```
def add_up():
    sum = 0
    try:
        while (True):
            num = int(input("Enter an int. Positive to sum, -1 to end"))
            if num < -1:
                raise ValueError
            elif num == -1:
                break
            else:
                sum = sum + num
    except:
        print("Enter a valid input")
    print(sum)
```

5) Write a function called `char_dict` that is passed a string and returns a dictionary that contains key-value pairs of individual characters in the string and the number of times which they occur. For example, the function call `char_dict("Snow, snow, go away!")` would return:

```
{'S': 1, 'n': 2, 'o': 3, 'w': 3, ',': 2, ' ': 3, 's': 1, 'g': 1, 'a': 2, 'y': 1, '!': 1}
```

```
def char_dict(s: str):
    d = dict()  or {}
    for x in range(0, len(s)):  or for c in text:
        c = s[x]
        val = d.get(c, 0)
        d.update({c: val+1})  or d[c] = val+1
    return d
```

6) You may recognize the following password validation function from Lecture 06.

```
def valid_password(password):
    correct_length = False
    has_uppercase = False
    has_lowercase = False
    has_digit = False

    if len(password) >= 7:
        correct_length = True

        for ch in password:
            if ch.isupper():
                has_uppercase = True
            if ch.islower():
                has_lowercase = True
            if ch.isdigit():
                has_digit = True

    return correct_length and has_uppercase and has_lowercase and has_digit
```

Write code that made the following modifications to the password:

(a) If the function returns False, concatenate `"abcABC123"` to the end of the password.

pass_bool = valid_password(t)
pass_bool = password + "abcABC123"

I don't set the 9.

(b) If the function returns False, for each uppercase character in the password, insert the corresponding lowercase character after it, and vise-a-versa. Concatenate the original length of the password to the end. For example: `"PASSwd"` would become `"PpAaSsSswWdD6"`. If the original length is less than 3 or if it contains no alphabetical characters, add `"Ab12"` to the front of it.

new pass = ""
for c in password:
if c.isupper():
newpass += c + c.lower()
elif c.islower():
newpass += c.upper()
return newpass + str(len(password))

if len(password) < 3:
return "Ab12" + password

check w/ code

if len(newpass) == 0:
return "Ab12" + password

(c) If the function returns True, reverse the order of the characters in the password.

password = password[::-1]

check if word too

password = password[::-1]